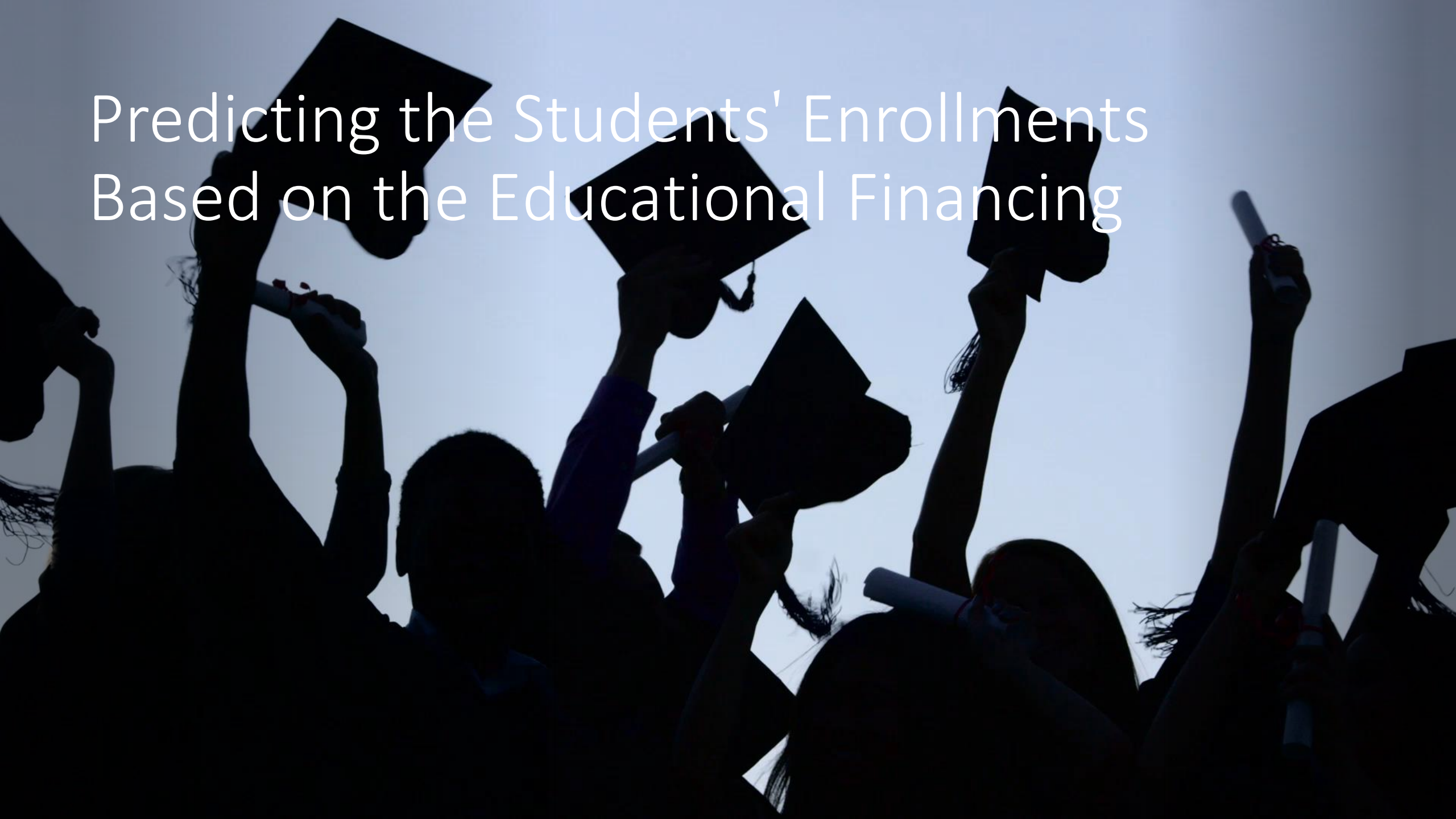


Predicting the Students' Enrollments Based on the Educational Financing



Statement of the Problems

- School district organization by strategic planning concerns the elementary student attrition and retention, and student enrollment is highly related to the financing management activities;
- The institutional support on the enrollment concerns that the secondary school student graduation college choice and the transition to colleges are related to the allocation of financial aid and considerable expenditure of awards.
- Considering the state and federal revenue and spending on instruction, this research explores different factors that presumably influence the enrollment amount of the school districts.
- This work intends to provide decision-makers in the enrollment management administration with a better understanding of the factors highly correlated to the enrollment capacity.
- We validate our methods using actual data from the NAEP (National Assessment of Educational Progress) and financing data from the National Finance Census (National Center for Educational Statistics).
- Linear regression (LR) and predict the enrollment of district and K-nearest Neighbor (KNN) on cohort levels.



Research Questions

What are the correlations between the features of educational financing and elementary and secondary students' enrollments in local school districts?



What is the model fit for the relevant educational financing features to predict the likelihood of enrollment in the next school year?



What is the recommendation to the stakeholders on financing management to increase the enrollment of elementary and secondary students?

Metrics and Feature Description

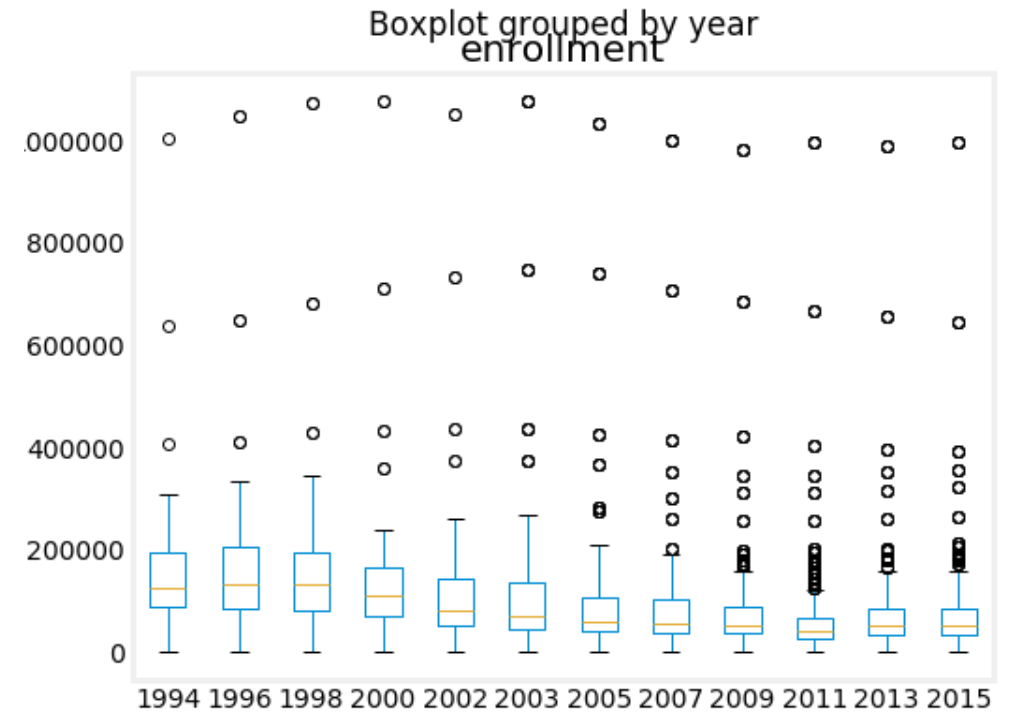
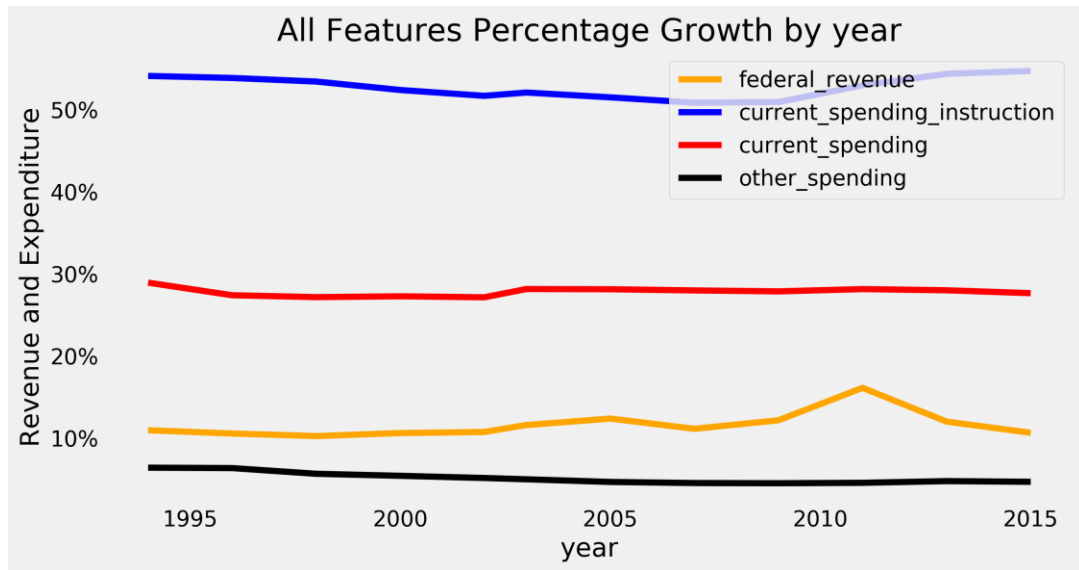
District Enrollment: overall enrollment number of elementary and secondary students aggregated in districts

Continuous: number count

Categorical: number count scale: < 50000, 50000 ~ 70000, > 70000

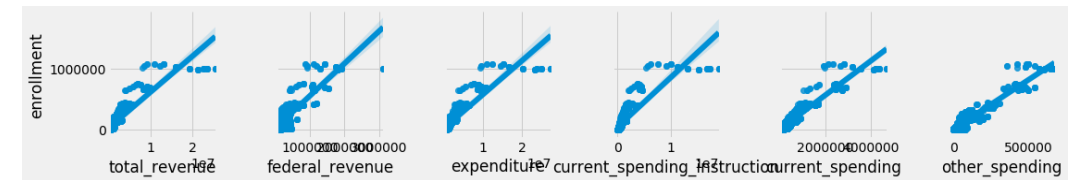
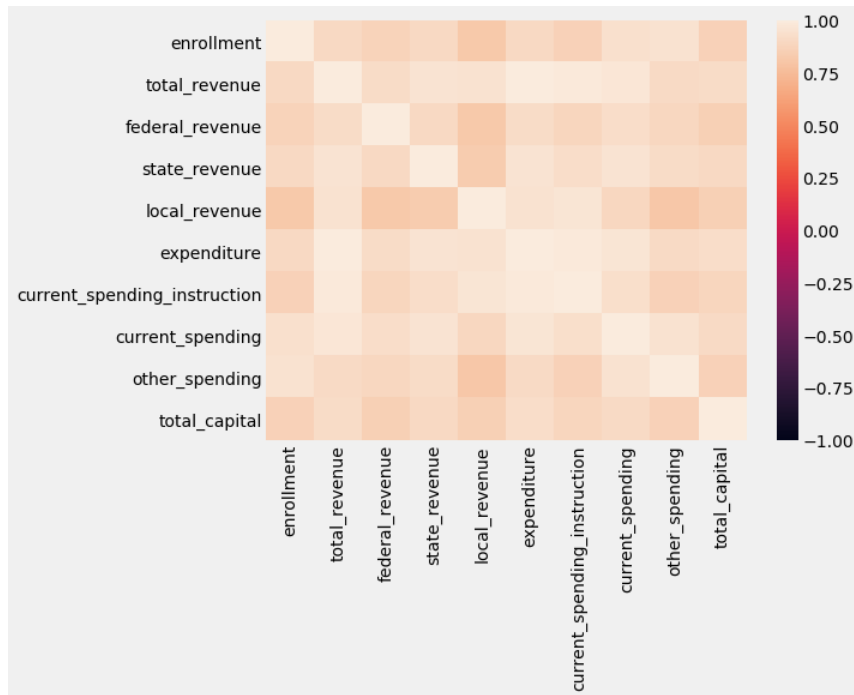
Features	Description	Feature Selection
Total elementary-secondary revenue	Sum of Revenue from Federal, State, Local Sources	*
Total Revenue from Federal Sources	• Compensatory (Title I) • federal aid	*
Total Revenue from State Sources	• General formula assistance • Special education programs • Transportation programs	
Total Revenue from Local Sources	• All taxes • Property taxes • Parent government contributions • Revenue from other school systems • Charges	*
Total elementary-secondary expenditure	Sum of current spending, payment to the government, total capitals.	*
Total Current Spending for Instruction	• Total salaries and wages • Total employee benefit payments	*
Other Current Spending	• Salaries and wages for instruction • Employee benefits for instruction	*
Total Capital Outlay Expenditure		

Education Financing Data Explorative Analysis



Features Correlation

- Features of revenue and spending have highly correlation to the enrollments on district levels.



Significant Predictors

```
=====
                        OLS Regression Results
=====
Dep. Variable:          enrollment    R-squared:                0.924
Model:                  OLS          Adj. R-squared:           0.924
Method:                 Least Squares  F-statistic:             7784.
Date:                   Sat, 13 Feb 2021  Prob (F-statistic):       0.00
Time:                   18:41:33      Log-Likelihood:          -45890.
No. Observations:      3855          AIC:                    9.179e+04
Df Residuals:          3848          BIC:                    9.184e+04
Df Model:               6
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                4230.3263     814.701        5.192     0.000     2633.039     5827.613
total_revenue         -0.0724         0.005    -13.166     0.000     -0.083     -0.062
federal_revenue       -0.0502         0.008     -6.654     0.000     -0.065     -0.035
expenditure           0.0493         0.006      8.693     0.000      0.038      0.060
current_spending_instruction  0.0159         0.006      2.781     0.005      0.005      0.027
current_spending       0.1727         0.010     16.700     0.000      0.152      0.193
other_spending         1.0255         0.027     38.685     0.000      0.974      1.077
=====
Omnibus:              2230.006    Durbin-Watson:           0.582
Prob(Omnibus):         0.000    Jarque-Bera (JB):       136071.575
Skew:                  1.988    Prob(JB):               0.00
Kurtosis:              31.833    Cond. No.               5.17e+06
=====
```

Warnings:

- Multiple Regression indicate the statistically significant predictors of the Features.
- The selected features have main effect on enrollment,
- $p < .001$

Performance Evaluation

LR Algorithm with Feature Selection

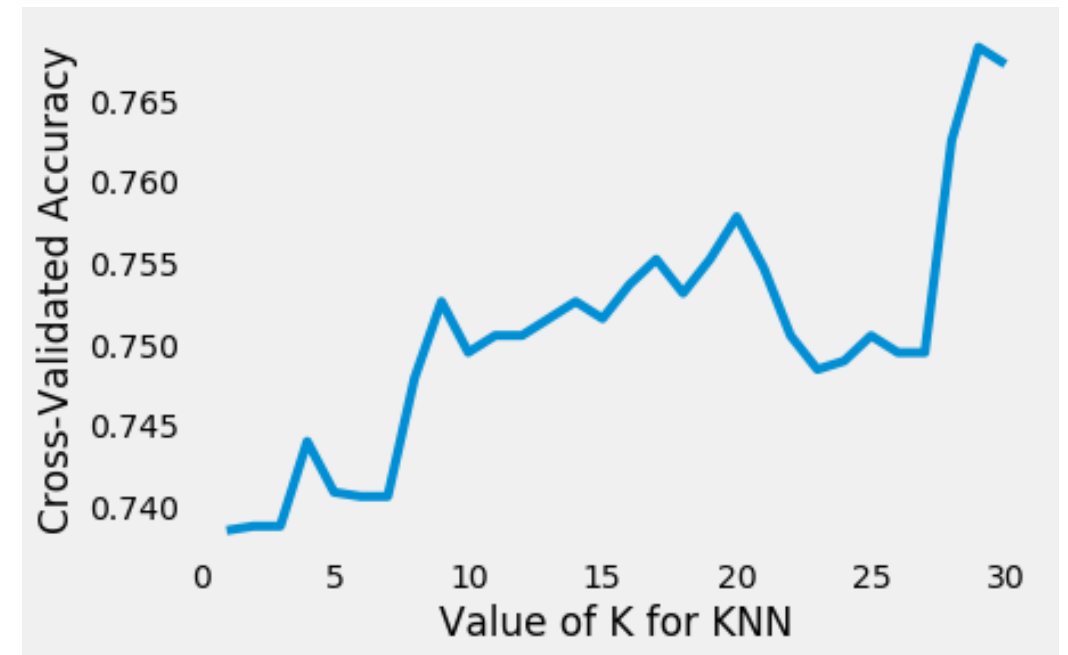
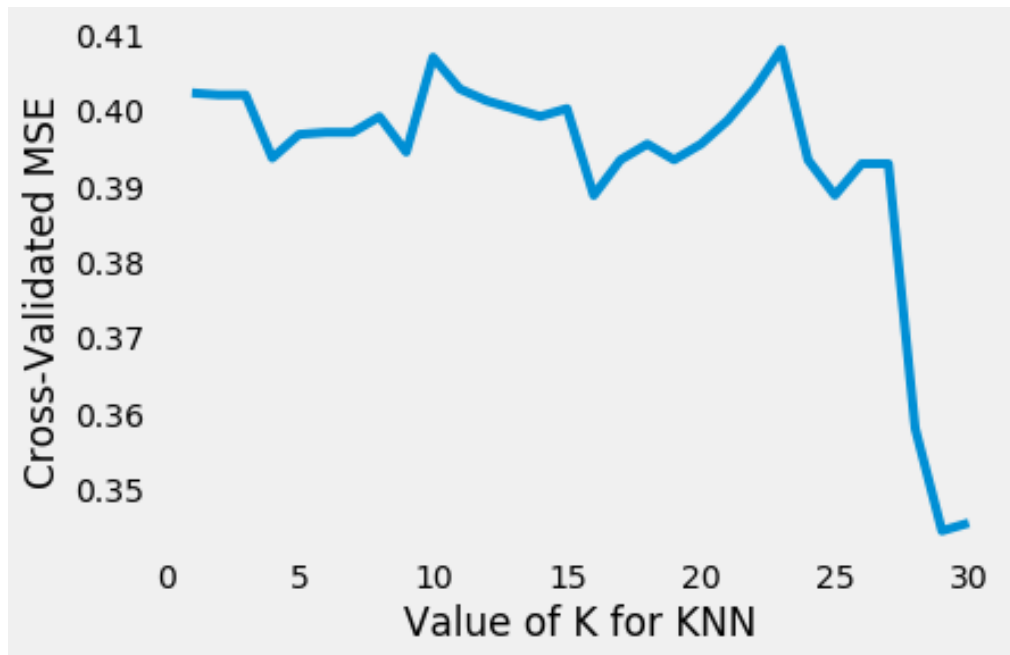
- $Enrollments = 4230.32 - 0.1 * total_revenue - 0.1 * federal_revenue + 0.1 * expenditure + 0.02 * current_spending_instruction + 0.2 * current_spending + 1.0 * other_spending$
- MAE: 19949.20
- RMSE: 30726.0

Regularized Regression Algorithm after compare the features

- $Enrollments = 4124 - 0.1 * total_revenue - 0.1 * federal_revenue + 0.16 * current_spending + .02 * current_spending_instruction + 1.1 * other_spending$
- RMSE: 30554.39 (alpha=0.0)

K Nearest Neighbors

- KNN is used to create the classifier to predict the enrollment with a result or roughly 93.14 % accuracy.
- 5-folds Cross-validation was used to evaluate the KNN algorithm
- Both the accuracy and MSE indicated the best K is 28 ~ 29



Comparison of LR & KNN

- The proposed LR algorithm and KNN models can predict enrollment with reliable accuracy.
- R Algorithm can be used to project next year's enrollment.

	LR	KNN
Train Data	2874	2891
Test Data	958	964
Accuracy	88.6%	93.14%

Findings & Implication

- The results indicated that spending on instruction, payments to the government, and institutional supports are more likely to increase district enrollments.
- Expanding the federal and total revenue, such as allocation of financial aid and scholarship, is not the best solution to maximize the enrollments for elementary and secondary students.
- Using the projection of the LR algorithm and KNN by applying the enrollment from last year to project next year's enrollment can help the management assign money rewards efficiently.



Recommendation for Stakeholders

- The elementary and secondary students' cohorts on the district level have considerable variation in managing educational financing and budget the next year compared to the school year.
- In planning and evaluating the enrollments, using the K-nearest neighbor would directly predict the cohort enrollment without identifying the variation in the school district's levels.
- For the research purpose, the institutional strategic planning would approach the enrollment by implementing LR projection.